Math 1610, Honors Calculus 1, Fall 2026, Homework 4

1. Suppose $S \subseteq \mathbb{R}$ is nonempty and bounded above, and let $s = \sup(S)$. Show that for every $\epsilon > 0$,

$$(s - \epsilon,\ s + \epsilon) \cap S \neq \emptyset.$$

Remember that it is <u>not</u> necessarily true that $s \in S$ — so you really do have to produce an element of S close to s, rather than simply pointing at s itself.

2. Suppose f is bounded and integrable on $[a-r, a+r]$ for some $r > 0$. Show that

$$\lim_{h \to 0} \int_a^{a+h} f(t)\, dt = 0.$$

(Note that h is allowed to be negative, in which case $\int_a^{a+h} f = -\int_{a+h}^{a} f$ by our convention.)

3. Throughout, A and B denote real numbers.

   (a) Show that $\big||A| - |B|\big| \leq |A - B|$.

   (b) Deduce that if $\lim_{x \to c} f(x) = L$, then $\lim_{x \to c} |f(x)| = |L|$.

(c) Show that

$$\max\{A, B\} = \frac{A + B + |A - B|}{2} \qquad \text{and} \qquad \min\{A, B\} = \frac{A + B - |A - B|}{2}.$$

(d) Deduce that if $\lim_{x \to c} f(x) = L$ and $\lim_{x \to c} g(x) = M$, then

$$\lim_{x \to c} \max\{f(x), g(x)\} = \max\{L, M\}.$$

4. (optional)

(a) Show that if $B_1, B_2 \neq 0$ then

$$\frac{A_1}{B_1} - \frac{A_2}{B_2} = \frac{A_1 B_2 - A_2 B_1}{B_1 B_2},$$

and in particular that $\dfrac{1}{A} - \dfrac{1}{B} = \dfrac{B - A}{AB}$ for $A, B \neq 0$.

(b) Suppose $\lim\limits_{x \to c} g(x) = M$ with $M \neq 0$. Show that

$$\lim_{x \to c} \frac{1}{g(x)} = \frac{1}{M}.$$

hint: the denominator in part (a) is not under your control, so get a lower bound for it first — show that $|g(x)| > |M|/2$ for all x close enough to c.